(Please write your Enrollment Number) Enrollment No. 194

MID-TERM EXAMINATION

Course Name: B. Tech. Semester: 5

(October, 2025) OFF LINE mode

Subject Code: BAI 301	Subject: Machine Learning
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) What are Data Visualization techniques?		CO1
	b) Explain the concept of K-Fold Cross Validation using example.		CO1
	c) Differentiate between Ridge Regression and LASSO Regression.		CO2
	d) Explain the working principle of Support Vector Machines. How do kernels help in handling non-linear data?		CO2
	UNIT I		CO Mapping
Q2	**Attempt any two parts**	**(5*2=10)**	**CO1**
	a) A binary classification model is evaluated on a test set of 190 samples. The confusion matrix obtained from the model's predictions is as follows:		

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	15	55

(i) Construct the confusion matrix and label its components (True Positives, True Negatives, False Positives, and False Negatives).

(ii) Calculate the following metrics for the model:

- Accuracy
- Precision
- Recall
- F1 Score

	b) Explain the importance of data cleaning and integration in machine learning. What are the common techniques used for handling missing or inconsistent data?		
	c) A researcher is working with a total dataset of N=25,000 geological samples.		

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	(i) 5-Fold Cross Validation (ii) Stratified 5-Fold Cross Validation Calculate how many samples will be in the training and testing sets for each fold.		
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	**CO2**
	a) Explain Decision Tree Learning with a neat diagram. Discuss its advantages and limitations.		
	b) The following dataset shows the study hours of students and their exam results (Pass = 1, Fail = 0):		

Student	Study Hours	Grade
S1	2	0
S2	4	0
S3	6	1
S4	8	1
S5	10	1

Classify a new student who studies for 5 hours using the KNN algorithm with k=3. Use Euclidean distance and show all steps.

c) Given a dataset with two features (X1 and X2) and a binary classification Label (A or B).

Data Point	X1	X2	Label
P1	2	6	A
P2	4	4	A
P3	5	8	B
P4	8	2	B
P5	10	7	A

Use the KNN algorithm with K=3 and the Euclidean Distance metric to classify a new data point P_{new} at coordinates (5,5).

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